

REVIEW-II

LexAgent

Multi-Agent Debate Framework for Hallucination Reduction in Legal AI

Student Name(s)

Guide

Programme / School

B.Tech, SENSE, VIT Chennai

Review & Date

Review-II · 30.09.2026 (Wednesday)

GUIDE'S SIGNATURE WITH DATE

Signature: _____

Date: _____

Problem and Objective

58—

88%

Hallucination rate of public LLMs on direct, verifiable questions about U.S. federal cases

GPT-4 58% · GPT-3.5 69% · PaLM 2 72% · Llama 2 88%

— Dahl et al., 2024 [12]

PROBLEM

Legal answers can contain invented cases, wrong citation identities and holdings that a real case does not support.

OBJECTIVE

Combine legal retrieval, adversarial debate and citation verification to reduce unsupported legal answers.

CURRENT SCOPE

U.S. federal case law from Harvard CAP. Prosecutor, Defense, Reflection and Judge share one 4-bit Mistral-7B on a single T4 GPU.

Objectives and Research Questions

#	OBJECTIVE	RESEARCH QUESTION
O1	Ground answers in legal sources	Does hybrid retrieval recover the target authority and the passage supporting its holding?
O2	Create an evidence-diverse debate	Does independent Defense retrieval reveal errors that shared-context debate misses?
O3	Verify each cited proposition	Can the verifier separate existence, support, fabrication and missing corpus coverage?
O4	Control debate effort	Does dynamic stopping improve quality per unit of latency over a fixed-round baseline?
O5	Evaluate reproducibly	Do gains persist on held-out cases with a fixed corpus, isolated memory and checked labels?

Literature Landscape: Five Method Families

FAMILY	PAPERS	CONTRIBUTION	GAP FOR LEGAL QA
Domain legal LLMs	Lawformer [1], DISC-LawLLM [8], LawGPT [16]	Legal fluency; DISC-LawLLM adds retrieval	Single generator; no independent citation veto
Legal benchmarks	LegalBench [7], LawBench [17], LexGLUE [3], LegalAgentBench [18]	Standard scoring of legal knowledge, reasoning and agents	No measure of citation refusal or re-hallucination
RAG and hybrid retrieval	RAG [2], CRAG [13], Self-RAG [10], LeAK-GraphRAG [20], AALawyer [21]	Grounds generation; corrects or constrains retrieval	Better evidence, but no adversarial check of the written claim
Detection and graph checks	Self-consistency [4], CoVe [9], HalluDetect [19], HalluGraph [23], Falkor-IRAC [25]	Entity and path checks beat embedding similarity	Post-hoc audit; no memory of past fabrications
Multi-agent debate	Du [5], MAD [6], Smit [11], Wang [14], Sun [15], ChatLaw [22], PROCLAIM [24]	Adversarial roles and progressive retrieval reduce false consensus	Configuration-sensitive; no legal citation matcher

Base paper [26] Meng et al., 2025: role-specialised human–AI legal workflow, which motivates LexAgent's role decomposition. Problem evidence: Dahl et al. [12].

Comparison of the Closest Systems

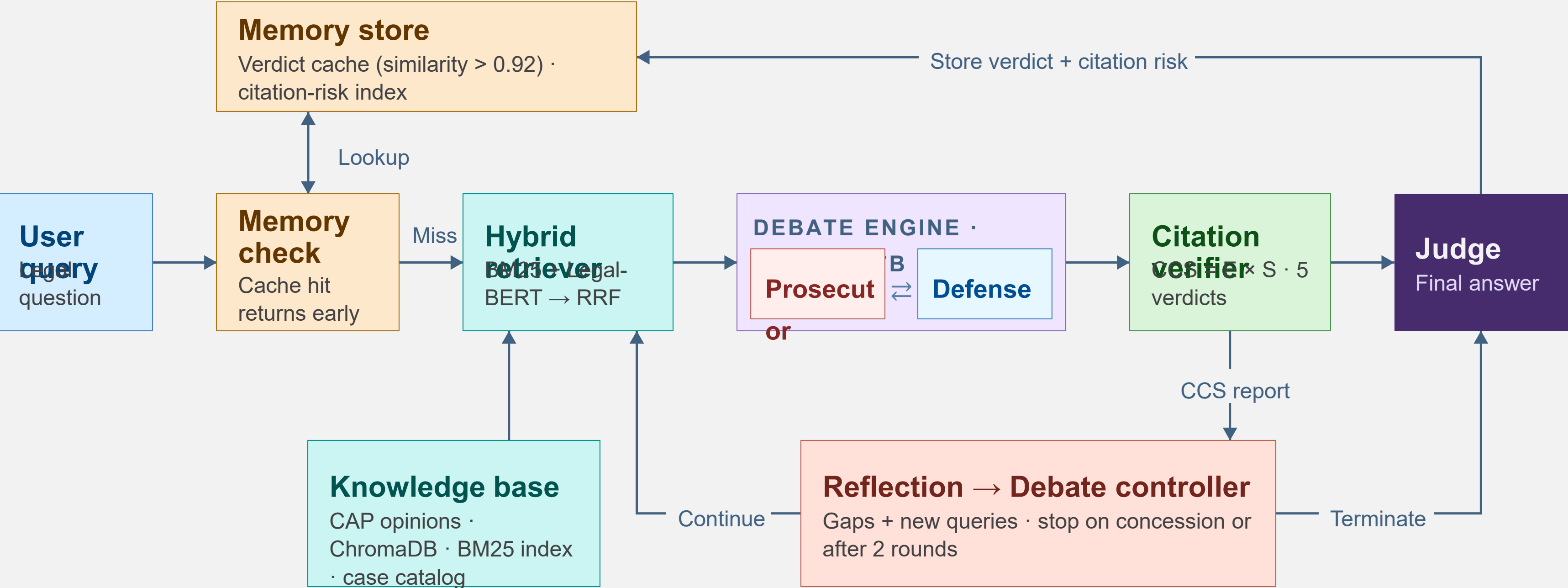
SYSTEM · DOMAIN	ARCHITECTURE	ADVERSARIAL DEBATE	IN-LOOP CITATION CHECK	HYBRID / STRUCTURED RETRIEVAL	CROSS-SESSION MEMORY
ChatLaw [22] Chinese law	RA-MoE, cooperative SOP	☐ No	☐ Semantic entropy	☒ Retrieval + KG	☐ No
HalluGraph [23] Legal RAG	Graph-alignment auditor	☐ No	☐ Post-generation	☐ No	☐ No
HalluDetect [19] Indian consumer law	Detector + cooperative agents	☐ No	☐ Post-generation	☒ RAG chatbots	☐ No
LeAK-GraphRAG [20] Chinese legal papers	GraphRAG	☐ No	☐ No	☒ Vector + Cypher	☐ No
AALawyer [21] Chinese criminal law	Dual RAG (SCR + APR)	☐ No	☐ Constrained retrieval	☒ Symbolic + analogical	☐ No
Falkor-IRAC [25] Indian judgments	IRAC graph + verifier veto	☐ Single generator	☒ Veto after proposal	☒ Symbolic graph	☐ No
PROCLAIM [24] Biomedical claims	Courtroom MAD + P-RAG	☒ Yes · $\approx 11\times$ tokens	☐ Evidence, not citations	☒ P-RAG only	☐ No
LexAgent (ours) U.S. federal case law	4-agent debate on one T4 GPU	☒ Evidence-disjoint	☒ $CCS = E \times S$	☒ BM25 + Legal-BERT	☒ Verdict + risk memory

☒ Ye ☒ Partial ☐ No In-loop check = corpus matching of citations during debate, not post-hoc scoring of a finished answer.

Research Gaps and LexAgent Response

GAP IN THE LITERATURE	LEXAGENT RESPONSE	VALIDATION STILL NEEDED
Retrieval can introduce misleading evidence [10, 13, 27]	Hybrid retrieval + reflection-driven query expansion	Recall of target cases and passages
Debate repeats shared errors [6, 11]	Defense counter-retrieval excludes Prosecutor evidence	Shared vs. disjoint evidence comparison
A real case may not support the claim [12, 23]	Separate case existence from holding support	Hard-negative stress tests, expert labels
Agreement and confidence can mislead [11, 12]	Citation-aware controller; Judge limited to verified list	Calibration and abstention
No system remembers rejected citations [24, 25]	Verdict cache + proposition-level risk memory	Paraphrase tests, cache isolation

System Architecture



Orchestrated as a LangGraph state machine. All agents share one 4-bit Mistral-7B on a single T4 GPU.

System Components

DATA	CAP case opinions 27 U.S. Reports volumes	Case catalog 14,524 name / reporter keys	Passage store 450-word chunks + metadata	Memory store Verdict cache + risk index
RETRIEVAL	BM25 lexical Names + terms · top 4	Legal-BERT dense ChromaDB · top 4	RRF fusion k = 60 · ≤ 8 passages	Defense counter- Retrieval Preserved docs excluded
AGENTS	Prosecutor Argument + citations	Defense Challenges + concession	Reflection Gaps + new queries	Judge Verified citations only
VERIFICATION	Existence · E Catalog + fuzzy name	Support · S DeBERTa NLI entailment	CCS = E × S Per-citation score	Five-way verdict Supported → Unverifiable
CONTROL	LangGraph state Query, evidence, history	Debate controller Stop or continue · ≤ 2 rounds	Mistral-7B · NF4 Shared by all roles · T4 GPU	Evaluation runners Benchmark + ablation

Debate Workflow



ORCHESTRATION

LangGraph state machine

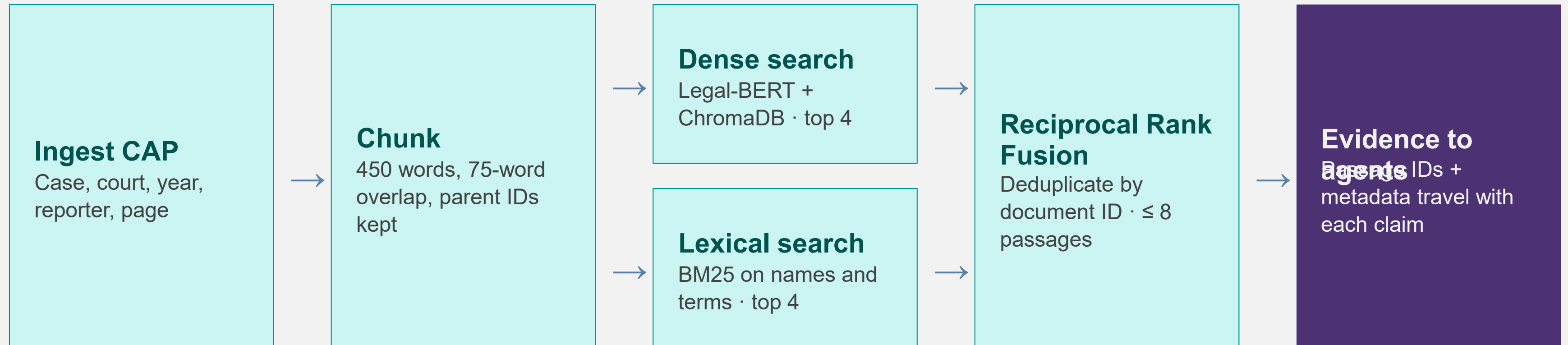
DEBATE BUDGET

Maximum two rounds per query

PROVENANCE

Passage IDs kept through debate and verification

Hybrid Retrieval Pipeline



$$RRF(d) = \sum \frac{1}{60 + rank_i(d)}$$

Exact case names and reporter keys come from BM25; paraphrased holdings come from dense search. Fusion rewards passages that rank well in both.

Agent Roles and Evidence-Disjoint Defense

Prosecutor Drafts an answer from retrieved case passages Argument + candidate citations	Defense Retrieves its own evidence and challenges the claims Challenges, counter-citations, concession	Reflection Reviews the debate and citation-confidence feedback Gaps, targeted queries, quality score	Judge Synthesises the debate; keeps only CVM-supported citations Answer, verdict, confidence
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Defense independence

COUNTER-QUERIES

Question + exception terms; cited case + “overruled”, “distinguished”

EVIDENCE EXCLUSION

Drop passages matching Prosecutor document IDs; keep up to four

EVIDENCE OVERLAP RATIO

Share of Defense citations found only in Prosecutor evidence. 1.0 = echo

All four roles share Mistral-7B-Instruct (4-bit). Role diversity comes from prompts and evidence.

Factorized Citation Verification

$$CCS = E \times S$$

E · Existence

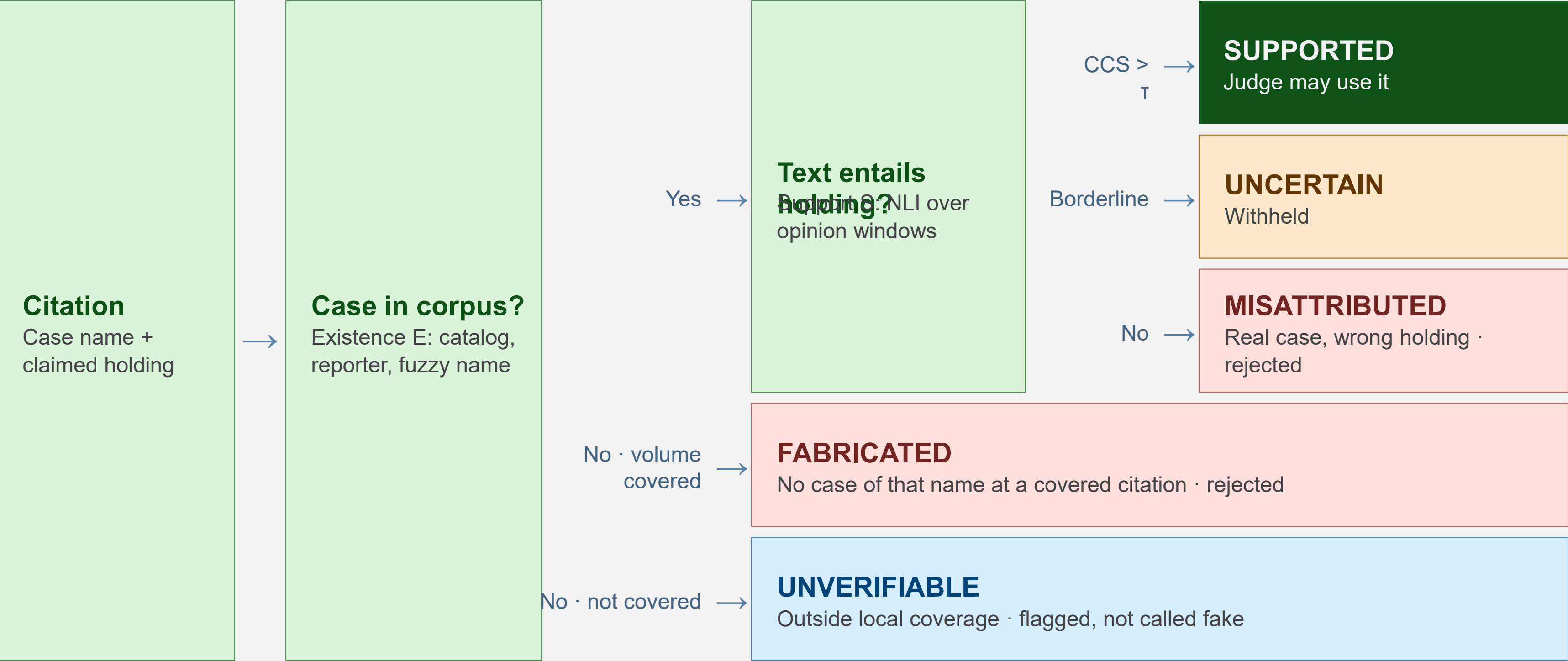
Exact catalog match, reporter/name check, fuzzy name match

S · Holding support

DeBERTa NLI cross-encoder: does the case's own text entail the claimed holding?

VERDICT	CONDITION	JUDGE MAY USE?
SUPPORTED	$E \geq 0.85$ and $S \geq 0.70$	Yes
UNCERTAIN	$E \geq 0.85$ and $0.40 \leq S < 0.70$	No
MISATTRIBUTED	$E \geq 0.85$ and $S < 0.40$	Rejected with reason
FABRICATED	Low E in a covered reporter volume	Rejected with reason
UNVERIFIABLE	Outside local corpus coverage	Flagged, not called fake

Citation Verdict Decision Flow



Proposed Novel Contributions

01 Factorized citation verification CCS = Existence × Support with a five-way verdict Replaces the weighted CCE sum	02 Conformal acceptance threshold $P = \frac{S_{(k)}, k}{ S_{(k)} + 1}(1 - \alpha)$, giving $P(\text{accept} \mid \text{unsupported}) \leq \alpha$ Replaces hand-picked 0.40 / 0.75 cut-offs	03 Evidence-disjoint Defense Defense retrieves its own counter-evidence; Prosecutor documents excluded Replaces shared-context Defense
04 Self-correcting memory Store (case, wrong holding) pairs; forget entries that later verify Replaces the name-level risk list	05 Citation stress-test protocol Inject labelled bad citations; measure catch rate per error type Replaces end-to-end RAGAS only	CONTRIBUTION STATEMENT LexAgent distinguishes fabricated, misattributed and unverifiable citations, forces evidential independence between debating agents, and bounds how often unsupported citations are accepted, all on one T4 GPU.

Implementation and Progress



LAYER	TECHNOLOGY	CONFIGURATION
Generation	Mistral-7B-Instruct-v0.3, NF4 4-bit	Tesla T4 16 GB · 768 max tokens · temp 0.1
Retrieval	Legal-BERT + ChromaDB, BM25, RRF (k = 60)	Top 4 + 4 → 8 passages · 450-word chunks · 27 volumes
Verification	DeBERTa NLI cross-encoder, factorized CVM	Per-citation identity and support reports
Orchestration	LangGraph, four roles, DDC, adaptive memory	Max two rounds · cache reuse at similarity > 0.92
Evaluation	Python CLI, Colab notebook, benchmark + ablation runners	25-case scorecard · 40-item stress test

Benchmark Results

25/25

Cases completed, no crashes

91.5%

Mean citation accuracy (CAS)

72%

Debate convergence (18/25)

19.4%

Mean hallucination rate (HR)

12 / 25

Cases with HR = 0

0.000

Evidence overlap ratio

73.2 min

One T4 GPU · ≈2.9 min per case

20 / 5

Two-round / one-round debates

READING THE HR

HR includes authentic Supreme Court citations outside the 27-volume local preset, such as Lockett v. Ohio and Milliken v. Bradley. These are unverifiable by this corpus, not confirmed fabrications.

Citation Stress Test

CATEGORY	EXPECTED	ACCEPTED BY VERIFIER		OUTCOME
Correct precedents (control)	Accept		10/10	As expected
Semantic misattributions	Reject		10/10	None caught
CaseHOLD legal distractors	Reject		5/5	Lexical overlap passes
Plausible fabricated citations	Reject		1/5	4 of 5 rejected
Fabricated, no citation details	Reject		0/5	All rejected
Real, off-corpus precedents	Flag		0/5	Unverifiable (coverage)



Component Ablation Study

VARIANT	MEAN HR		CAS	CONFID.
Full · DVA+ Complete LexAgent v3.0		0.168	0.911	0.763
A1 · No Defense Monologue baseline		0.000	0.906	0.766
A2 · No CCE Citation verification removed		0.133	0.914	0.700
A3 · No memory Adaptive memory removed		0.225	0.926	0.865
A4 · No DDC Fixed one round		0.058	0.909	0.780

+33.9%

Relative HR increase without adaptive memory (0.168 → 0.225)

CAUTION

A1's zero HR comes from removing the adversary, not from stronger robustness.

CONFIDENCE

A3 has the highest HR and the highest confidence, so confidence alone is not reliability.

Worked Example

QUERY

Does the Fourth Amendment protect against warrantless cell phone searches?

PHASE	SAVED OBSERVATION	WHAT IT SHOWS
1 · Hybrid retrieval	8 CAP authority passages returned	Dense and lexical retrieval work together
2 · Fixed one-round debate	Riley citation rejected as unverified; Judge confidence 0.35	An answer can outlive a failed citation check
3 · Full DVA+	2 rounds, average CCS 0.998, one verified citation, confidence 1.00	Verification and stopping state are recorded

The separate benchmark run reports 66.67% HR for this case, so versions will be pinned and trials repeated.

Remaining Work and Timeline

10%

Verifier + final-answer checks

8%

Baselines + ablations

4%

Memory + reproducibility

3%

Final report + demo

30 Sep – 6 Oct

Audit identity resolution, NLI mapping, final-answer checks
Verifier tests, revised stress test

7 – 13 Oct

Matched RAG baseline and four ablations
Comparable scorecards, latency, tokens

14 – 20 Oct

Held-out review, paraphrase and memory tests
Independent labels, failure taxonomy

21 – 27 Oct

Repeat experiments; finalise report and demo
Reproducible package, final figures

28 Oct

Review-III
Final results and limitations

Conclusions

Prototype achieved

CAP retrieval, four-role debate, factorized citation verification, dynamic stopping and adaptive memory run end-to-end on one T4.

Open issues

Semantic misattributions and CaseHOLD distractors pass the verifier; off-corpus citations inflate the measured HR.

Next milestone

Repair holding-support checks, calibrate τ conformally, then test against a matched RAG baseline and ablations.

Thank
you

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